

When Determinants Are Not Enough: Private Rare Switching

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Abstract

In this post, I would like to share a small research moment where Codex helped me find the right way to adapt rare switching to the private setting. The standard determinant-based update rule in linear bandits and RL works beautifully because the design matrix grows monotonically. But once Gaussian noise is added for privacy, this monotonicity can fail, and the usual analysis no longer goes through. The key reason is that determinant growth controls volume, while regret analysis needs control of the worst direction. To address this, Codex comes up with a different rare-switching rule based on the generalized Rayleigh quotient, which restores logarithmic policy updates and the desired confidence-width comparison up to a constant factor.

1 The Problem

The problem is a variant of the standard rare switching policy update in linear bandit and RL. Let us first recall the standard setting, i.e., Sec. 5.1 in [APS11]. The learner maintains a positive-definite covariance matrix $V_t = \lambda I + \sum_{s=1}^t x_s x_s^\top$ for $t \in [T]$, $\lambda > 0$ and only updates its policy when $\det(V_t) > \alpha \det(V_\tau)$ for some $\alpha > 1$ with τ being the latest update time slot before t . This simple rule is often called *rare-switching update* which can (i) reduce the number of total policy updates to $O(\log T)$ (which directly follows from the standard elliptical potential lemma, Lemma 11 in [APS11]) while (ii) only suffers a constant factor blow up in regret (which directly follows from Lemma 12 in [APS11]). In particular, Lemma 12 in [APS11] implies that if $V_t \succeq V_\tau$, then we have for any $x \in \mathbb{R}^d$, we have $\|x\|_{V_\tau^{-1}} \leq \sqrt{\alpha} \|x\|_{V_t^{-1}}$.

Now, what if the covariance matrix is perturbed, say, due to privacy protection. For example, in private linear contextual bandits or linear RL, Gaussian noisy matrix is often added to the non-private to arrive at the private one: $\tilde{V}_t = V_t + E_t$, where E_t is symmetric matrix with elements sampled from Gaussian. A natural question to ask is: Can we still apply the above update rule (on $\tilde{V}_t$) along with its analysis to arrive at the same performance, i.e., log-order updates with the same-order of regret?

The answer is no. The reason is that to apply Lemma 12, we have to guarantee the monotonicity property, which is $\tilde{V}_t \succeq \tilde{V}_\tau$. This is no longer true due to the private noise. This issue has been pointed by my previous work [CZ22] (cf. Sec. 6) and also by my most recent work [HZ26] (cf. Appendix C).

So, we may ask—how about a new rare-switching rule for the private case¹?

2 The New Rule

I basically present the above write-up as a prompt to Codex 5.5 (Extra High) with all related papers in the folder. The first attempt of Codex gives me the correct answer, which is summarized in the following

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¹Of course, one may ask if we can keep the same update rule but with a more advanced analysis to maintain the same performance guarantees. It turns out that this is not possible, which will become clear at the end of this post.

theorem².

Theorem 2.1. Let $x_1, \dots, x_T \in \mathbb{R}^d$ satisfy $\|x_t\|_2 \leq L$ for all $t \in [T]$. Let the non-private design matrices be

$$V_1 := \lambda I, V_{t+1} := V_t + x_t x_t^\top.$$

Assume the private matrices be $\tilde{V}_t = V_t + E_t$ where E_t is a symmetric matrix such that for all $t \in [T]$ ³

$$\|E_t\|_2 \leq \eta \text{ and } 0 \leq 2\eta \leq \lambda.$$

Suppose that the update rule is that for any $t \in [T]$, it updates the policy when

$$\lambda_{\max} \left(\tilde{V}_{\tau_t}^{-1/2} \tilde{V}_t \tilde{V}_{\tau_t}^{-1/2} \right) > \alpha,$$

where τ_t is again the most recent update before t .

Then, let $\rho := \eta/\lambda$ and $c_\rho := \frac{1+\rho}{1-\rho}$, we have

(i) For any $x \in \mathbb{R}^d$ and $t \in [T]$, $\|x\|_{\tilde{V}_t^{-1}} \leq \sqrt{\alpha} \|x\|_{\tilde{V}_{\tau_t}^{-1}}$ and $\|x\|_{V_t^{-1}} \leq \sqrt{c_\rho \alpha} \|x\|_{V_{\tau_t}^{-1}}$;

(ii) The total number of updates m is upper bounded by $m \lesssim d \log_{\alpha/c_\rho} \left(1 + \frac{TL^2}{\lambda d} \right) = O_d(\log T)$.

Remark 2.2. Several remarks are in order. First, in most cases, λ is set to be 2η so that the private design matrix is also positive definite, see the proof of Lemma A.4 in [CZ22]. Hence, $\rho = 1/2$ and $c_\rho = 3$. Now, with the above update rule and guarantees, we have the right way to do rare switching in private linear bandit and RL. Second, note that we often only needs the first inequality in (i) to bound the regret. Finally, we also note that the new update rule is also equivalent to that the learner updates iff $\tilde{V}_t \not\preceq \alpha \tilde{V}_{\tau_t}$, which is more expensive in computation compared with the old determinant-based update rule (which can leverage the efficient rank-one update).

Proof. We first present the following claim, which will be useful for the proof with the proof being given at the end.

Claim 2.3. For $1 \leq a \leq b \leq T$, let $g_{a,b} := \lambda_{\max} \left(V_a^{-1/2} V_b V_a^{-1/2} \right)$ and $\tilde{g}_{a,b} := \lambda_{\max} \left(\tilde{V}_a^{-1/2} \tilde{V}_b \tilde{V}_a^{-1/2} \right)$. Then, we have $c_\rho^{-1} g_{a,b} \leq \tilde{g}_{a,b} \leq c_\rho g_{a,b}$.

We start with (i). For any $\tau_t < t$, i.e., t is not the update slot, the first inequality follows directly from the update rule. For the second one, by Claim 2.3 and our update rule, we first have

$$\lambda_{\max} \left(V_{\tau_t}^{-1/2} V_t V_{\tau_t}^{-1/2} \right) \leq c_\rho \alpha.$$

Further, by the identity of generalized Rayleigh quotient, we have

$$\sup_{x \neq 0} \frac{x^\top V_t x}{x^\top V_{\tau_t} x} = \lambda_{\max} \left(V_{\tau_t}^{-1/2} V_t V_{\tau_t}^{-1/2} \right) \leq c_\rho \alpha.$$

This implies that $V_{\tau_t}^{-1} \preceq c_\rho \alpha V_t^{-1}$, and hence $\|x\|_{V_{\tau_t}^{-1}} \leq \sqrt{c_\rho \alpha} \|x\|_{V_t^{-1}}$.

²I clean up and re-organize the original proof of Codex, which is correct but not that easy to read.

³For Gaussian matrix, this condition holds with high probability

We now move to (ii). Consider two adjacent policy update slots $i < j$. By the update rule and Claim 2.3, we have

$$\lambda_{\max} \left(V_i^{-1/2} V_j V_i^{-1/2} \right) > \alpha / c_\rho.$$

Further, since $V_j \succeq V_i$, all eigenvalues of $V_i^{-1/2} V_j V_i^{-1/2}$ are least one. Hence, by the above result and multiplicativity of determinant, we have

$$\frac{\det(V_j)}{\det(V_i)} = \det \left(V_i^{-1/2} V_j V_i^{-1/2} \right) > \alpha / c_\rho.$$

Thus, by the elliptical potential lemma (cf. Lemma 11 in [APS11]), we have that the total number of updates m is upper bounded by $m \lesssim d \log_{\alpha/c_\rho} \left(1 + \frac{TL^2}{\lambda d} \right)$.

We are left with the proof of Claim 2.3. To show it, we first notice that by the condition on E_t , we have

$$-\eta I \preceq E_t \preceq \eta I.$$

Further, by $V_t \succeq \lambda I$, hence $\eta I \preceq (\eta/\lambda) V_t = \rho V_t$. Combing the above, yields that

$$(1 - \rho) V_t \preceq \tilde{V}_t \preceq (1 + \rho) V_t.$$

Thus, for any nonzero vector $x \in \mathbb{R}^d$, we have

$$\frac{x^\top \tilde{V}_b x}{x^\top \tilde{V}_a x} \leq \frac{(1 + \rho) x^\top V_b x}{(1 - \rho) x^\top V_a x} = c_\rho \frac{x^\top V_b x}{x^\top V_a x}.$$

Taking sup and applying identity of generalized Rayleigh quotient, yields that $\tilde{g}_{a,b} \leq c_\rho g_{a,b}$. The other direction follows the same proof. $\square$

3 The Connection

One may wonder what's the difference or connection between the new update rule and the standard one. To see this, it is instructive to see the behavior of the new update rule even in the non-private case and ask whether it leads to a more frequent or less frequent update. To this end, I directly prompt Codex (also with ChatGPT) with the the screenshot of Lemma 12 in [APS11]. It then gives me a new proof which actually sheds a lot of light on how Codex comes up with the new update rule and its connection with the old one. The new proof is much simpler (at least to me) compared with the original proof. Basically, the proof first notes that LHS in Lemma 12 is equal to

$$\sup_{x \neq 0} \frac{x^\top A x}{x^\top B x} = \lambda_{\max}(B^{-1/2} A B^{-1/2}), \quad (1)$$

where again we apply identity of generalized Rayleigh quotient, while the RHS in Lemma 12 is equal to

$$\frac{\det(A)}{\det(B)} = \det(B^{-1/2} A B^{-1/2}). \quad (2)$$

Note that (1) $\leq$ (2) by the fact that all eigenvalues of $B^{-1/2} A B^{-1/2}$ are at least one, which comes from the monotonicity that $B \preceq A$. This proves Lemma 12.

So, essentially, the new update rule comes from the new proof of Lemma 12, i.e., (1). Moreover, we can see that under the new update rule, the learner would update less frequent as (1) $> \alpha$ implies (2) $> \alpha$, but not the other direction.

This new proof also explains why the new update rule can get rid of the issue in the old rule when lacking of monotonicity. This is because using the old update rule (cf. (2)), one has to further leverage the monotonicity to arrive at the bound on $\sup_{x \neq 0} \frac{x^\top Ax}{x^\top Bx}$, i.e., the worst-case direction. On the other hand, the new rule directly works with $\sup_{x \neq 0} \frac{x^\top Ax}{x^\top Bx}$.

We can now also see that with the old update rule, it is impossible to control the worst case direction without monotonicity since the eigenvalues on other directions can be much smaller and hence a very large value on the worst-case direction can still make the determinant small.

References

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